

Hyunwon Chung

hyunwon@umich.edu

INTERESTS	Parallel Computing, Reconfigurable architecture, CGRA (Coarse-Grained Reconfigurable Architecture), Dataflow Architecture, Low-power SoC, VLSI design	
EDUCATION	University of Michigan , Ann Arbor <i>MS, PhD. in Electrical and Computer Engineering</i>	Aug 2022 – Current
	Korea University , Seoul <i>BS. in Electrical Engineering</i>	Mar 2015 – Feb 2022 (Include 2-year military service)
RESEARCH	PROWESS (Processor Reconfiguration for Wideband Sensor Systems)	
	<i>Co-advisors: Prof. David Blaauw and Prof. Hun-Seok Kim</i>	University of Michigan
	• Architected streaming Processor with a hardware kernel launcher enabling 10ns runtime reconfiguration without CPU intervention • Designed a dynamic scheduling unit for graph-based, data-dependent kernel execution under multi-channel RF workloads • Achieved 248.5 GFLOPS/mm² compute density and 1.67 TFLOPS/W energy efficiency in 28nm CMOS • Delivered up to 8.3× end-to-end performance improvement and 5× PE utilization via concurrent execution and speculative preloading	
	COCHON (Configurable Optical Communications via Heterogeneous-Processing Optimized Node)	
	<i>Co-advisors: Prof. David Blaauw and Prof. Hun-Seok Kim</i>	University of Michigan
	• Contributed to a 130 mm² heterogeneous processor integrating both optical transceiver and receiver side digital baseband for beyond-5G communication • Led top-level physical design focusing on interface timing and clock-domain crossing among fully asynchronous cores • Delivered end-to-end silicon using GF 12 nm process, coordinating across multiple accelerators, and SoC teams	
	DASH-SoC (Domain-Focused Advanced Software Reconfiguration Heterogeneous)	
	<i>Co-advisors: Prof. David Blaauw and Prof. Hun-Seok Kim</i>	University of Michigan
	• Contributed to the development of two accelerators as part of the DARPA-funded DASH project • Delivered GF 12 nm tape-out as part of the DARPA DASH accelerator family • DAP (Domain Adaptive Processor) • Improved the DAP compute architecture from 16-bit to 32-bit, achieving substantial performance gain • Enabled multi-kernel execution to support diverse communication workloads • Overhauled the physical design, improving congestion, timing closure, and overall silicon efficiency • Fully Reconfigurable Accelerator for FEC • Designed a flexible forward-error-correction processor supporting 3GPP LTE/5G, WiFi, and SDA OCT standards • Integrated six major FEC modes into a single architecture, providing high adaptability with minimal area overhead	
	Automatic Peak Detection Algorithm based on Continuous Wavelet Transform	
	<i>Advisors: Prof. Yogesh Gianchandani</i>	University of Michigan
	• Developed software and algorithm pipeline for multi-detector micro-GC chromatogram analysis • Improved peak-detection and baseline-correction accuracy using CWT-based sparse feature extraction • Co-authored publication accepted to Journal of Chromatography A (2024)	
	Low Power Accelerators for ML & Communication Systems	
	<i>Advisor: Prof. Jongsun Park</i>	Korea University

- Designed a DCT-based JPEG encoder/decoder architecture and synthesized the RTL in 65nm TSMC Technology
- Developed a low-power hardware implementation of a Viterbi decoder architecture
- Proposed an 8-bit fixed-point low-power accelerator for convolutional neural networks
- Designed a folded architecture for fully-connected layers to improve compute efficiency

PUBLICATION

Yue, Yufan, Kuan-Yu Chen, Xiangdoing Wei, Tutu Ajayi, **Hyunwon Chung**, Ronald Dreslinski, David Blaauw, Hun-Seok Kim. "QFEC: A 9.97Gb/s Fully Configurable Quad-Mode Decoder for LDPC, Polar, Turbo, and Convolutional Codes." IEEE Transactions on Circuits and Systems I: Regular Papers (In-Progress)

Lin, Jiahao, H. Umut Suluhan, **Hyunwon Chung**, Arindam Dutta, Anish Vipplerla, Gerard Gubash, Jacob Holtom et al. "An Overview of Challenges and Requirements for Real-Time Spectrum Sensing in Modern RF Autonomy Systems." IEEE Design & Test (2025)

Zhao, Xiangyu, Ryan Aridi, Jacob Hume, Swetha Subbiah, Xingqi Wu, **Hyunwon Chung**, Yutao Qin, and Yogesh B. Gianchandani. "Automatic peak detection algorithm based on continuous wavelet transform for complex chromatograms from multi-detector micro-scale gas chromatographs." Journal of Chromatography A (2024)

SELECTED PROJECTS

A Dynamic Quantized CNN Processor with Analog MAC Operations

- EECS 627 course project at the University of Michigan
- Designed analog MAC unit which includes digital to pulse converter, analog datapath, and flash ADC
- Proposed dynamic quantization method which allows each layer have different fixed points
- Post-quantization training and quantization-aware training are both used for 5-bit quantization, to get proper accuracy on MNIST and Cifar-10 dataset

16 bit Microprocessor with In-memory computing technology based on 6T SRAM architecture

- EECS 427 course project at the University of Michigan
- Developed 16-bit Microprocessor based on RISC-V ISA, which includes RF, ALU, Controller, Program Counter, Shifter, and IMEM/DMEM
- Designed customized 6T-SRAM based in-memory computing (IMC) module into 16-bit microprocessor

Custom Design Projects

- Implemented 128-bit Lightweight Encryption Algorithm (LEA) Decryption security module using Matlab & Verilog and synthesized it using 90nm TSMC technology
- Developed several circuit design including Inverter Chain, Ring Oscillator, SRAM, 3-tap FIR filter, and Flip-Flops. This includes the schematic, layout, and post-layout simulation

The World Embedded Software Contest - AI Humanoid Division

- Developed a program for real-time image processing and humanoid robot control using python
- Won 3rd place and Awarded Embedded SW & System Industry Association Chairman's award

Computer Architecture Project

- Developed Inverse-matrix program and heap-sort program using ARM assembly language
- Implemented TCP Server-Client program using multi-thread technique in C

SKILLS

Languages : C, C++, Verilog, SystemVerilog, Python, OpenMP, CUDA, PyTorch, Tensorflow
Hardware Simulation Tool : PSPICE, Design Compiler, Model Sim, VCS
Circuit Design Software : Virtuoso, Innovus, Calibre

MILITARY SERVICE

Korean Augmentation To the United States Army,
Squad Leader, Sergeant, Headquarter, 23rd CBERN, 2ID, Camp Humphreys Sep 2016 – Jun 2018

AWARDS & SCHOLARSHIPS

Electrical and Computer Engineering Department Graduate Fellowship, University of Michigan	2025
Dean's list at School of Electrical Engineering, Korea University	2018 - 2021
Miraero Scholarship at School of Electrical Engineering, Korea University	2016
National Scholarship (Admission with highest distinction), Korea University	2015